

SMART INDIA HACKATHON 2026

TITLE PAGE

Problem Statement ID - 26153

**Problem Statement Title - AI-based Network Attack
Forecasting from Network Traffic Data using World
Models**

Theme - <exact theme from SIH portal>

PS Category - Software

Team ID - <from portal>

**Team Name (Registered on portal) - <registered
team name>**

**ARGUS — an offline AI that predicts
the next attack stage BEFORE the
hacker lands, and explains why.**

Autonomous Attack Forecasting & Multi-Agent SOC Co-Pilot

CORE DOMAIN SKILLS

PyTorch

FastAPI

Docker

ATT&CK

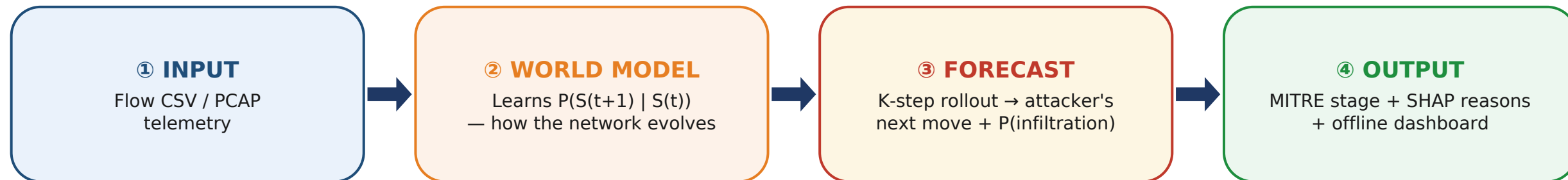
SHAP

Open Source

IDEA TITLE

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

ARGUS — Autonomous Attack Forecasting & Multi-Agent SOC Co-Pilot



✓ Detailed explanation

- Watches traffic as a time-story, not isolated packets
- Learns network world model: $P(\text{next state} | \text{current state})$
- Forecasts K steps ahead → attack probability before compromise
- Maps forecast to MITRE ATT&CK kill-chain stages
- Explains every alert: SHAP + attention weights
- Runs 100% offline — no cloud APIs, air-gap ready

✓ How it addresses the problem

- Today's IDS alarms AFTER entry — 241 days avg to contain (IBM 2025)
- ARGUS warns during recon / lateral movement — before exfiltration

✓ Innovation & uniqueness

- 1st open-source combo: real flow+packet telemetry → learned state transitions → K-step attack forecast
- Dual engine: Temporal Transformer + zero-dependency Markov fallback
- Auto-generates CERT-In (6-hr SLA) & NCIIPC CII advisories

Your
Team Name

TECHNICAL APPROACH

◆ Technologies to be used · ◆ Methodology & process (flow) · ◆ Working prototype

Languages & Core: Python 3.10 · pandas · Scapy / PyShark

ML / AI Engine: PyTorch Temporal Transformer (4-layer, 612K params) · Random Forest + TreeSHAP · Markov fallback

Backend & Logic: FastAPI · SSE streaming · MCP tool server · CLI

Frontend / UI: Offline single-page SOC dashboard (HTML/JS) — no internet

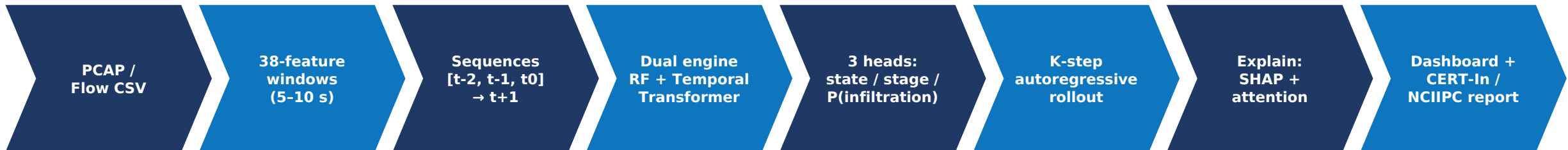
Data — 8 loaders: CIC-IDS-2017/18 · UNSW-NB15 · CTU-13 · CICIoT2023 · LANL · DARPA/NSL-KDD · synthetic

Knowledge & Standards: MITRE ATT&CK v14 (697 techniques) · CAPEC · NIST NVD · NetFlow / PCAP / IPFIX

Deploy / Hardware: Docker · CPU-only (no GPU) · fully air-gapped capable

✓ **Working Prototype:** 5-tab dashboard · CLI skills · 37/37 tests green · 1-command docker run

METHODOLOGY — INPUT → PROCESS → OUTPUT FLOW



Temporal split — no future leakage · Combined loss: $MSE(state) + CE(stage) + BCE(infiltration) = 0.3 / 0.4 / 0.3$



FEASIBILITY AND VIABILITY

◆ Analysis of feasibility · ◆ Potential challenges and risks · ◆ Strategies for overcoming these challenges

ANALYSIS OF FEASIBILITY

- Technical: 612K-param model trains in ~58 s on CPU; real-time inference
- Operational: drop-in on existing NetFlow/PCAP sensors; Docker = 1 command
- Economic: 100% open-source → zero cloud / API / licence cost
- Data: 8 public benchmark datasets already bundled & normalized
- Already prototyped: live dashboard + CLI + 37/37 tests green

POTENTIAL CHALLENGES & RISKS

- False alarms / alert fatigue on never-seen attack variants
- Raw PCAP volume (GBs) & class imbalance in labels
- Model drift as attackers change tactics (TTPs)
- AI co-pilot itself: prompt-injection & PII-leakage risk
- Legacy / air-gapped CII networks with no GPU

STRATEGIES TO OVERCOME

- Temporal no-leak splits + 8-dataset eval vs LR/RF baseline (FPR per class)
- Flow aggregation before inference — never feed raw PCAP to the model
- Continuous retraining triggers + kill-chain transition monitoring
- Guardrails: PII redaction · RBAC · injection scan · HMAC audit · human approval
- Markov fallback — zero ML dependencies on old hardware

SECURITY & COMPLIANCE

Air-gapped / no cloud APIs · AES-256 at rest · tamper-evident SHA-256 HMAC audit · CERT-In 6-hour SLA (Sec 70B) · NCIIPC CII advisory (Sec 70A — 6 sectors)

IMPACT AND BENEFITS

◆ Measurable impact (benchmarked) · ◆ Potential impact on the target audience · ◆ Benefits (social, economic, environmental)

99.97%

World-model accuracy
(vs 93.0% RF baseline)

0.983

Infiltration probability
AUC-ROC

K = 5

windows of early warning
before compromise

8

public benchmark
datasets supported

37/37

automated tests
green

POTENTIAL IMPACT ON THE TARGET AUDIENCE

- NTRO / NCIIPC: pre-breach forecasting for 6 CII sectors — power, BFSI, telecom, transport, govt, defence
- SOC analysts: a few ranked, explained alerts — not thousands of noisy late alarms
- Enterprises / ISPs / hospitals / railways: affordable, offline, Indian-made defence stack
- Students & open-source community: extensible MIT-licensed base for Indian cyber research

BENEFITS OF THE SOLUTION (SOCIAL · ECONOMIC · STRATEGIC)

- Social: protects hospital, power-grid & citizen data → trust in Digital India
- Economic: breach avg \$4.44M & 241-day containment (IBM 2025) — ARGUS shortens detect→contain
- National / strategic: Atmanirbhar sovereign offline stack — no foreign cloud, telemetry never leaves
- Environmental: CPU-only inference, no GPU farm · Ecosystem: open-source SOC tooling for India



RESEARCH AND REFERENCES

◆ Details / Links of the reference and research work

PROBLEM STATEMENT & STATUTORY REFERENCES

- SIH PS 26153 — AI-based Network Attack Forecasting using World Models (NTRO / NCIIPC), SIH 2026
- CERT-In Cyber Security Directions 2022 — 6-hour incident reporting — <https://www.cert-in.org.in>
- NCIIPC — Sec 70A, IT Act 2000 — CII protection — <https://nciipc.gov.in>
- MITRE ATT&CK v14 — <https://attack.mitre.org> · CAPEC · NIST NVD API 2.0 (CVSS v3.1)
- IBM Cost of a Data Breach Report 2025 — \$4.44M avg · 241-day containment — <https://ibm.com/reports/data-breach>
- Datasets: CIC-IDS-2017/18 — <https://www.unb.ca/cic/datasets/ids-2017.html>

RESEARCH FOUNDATIONS (RELATED WORK)

- Field survey — Husák, Komárková, Bou-Harb, Čeleda: “Survey of Attack Projection, Prediction and Forecasting in Cyber Security”, IEEE COMST 2019
- World models — Ha & Schmidhuber 2018, arxiv.org/abs/1803.10122 · DreamerV3, Hafner et al. 2023
- Security sequence prediction — Tiresias (CCS 2018) · DeepLog (CCS 2017) · DeepCASE (IEEE S&P 2022)
- Graph / provenance NIDS — E-GraphSAGE · Anomal-E · Kairos (IEEE S&P 2024)
- Explainability — SHAP: Lundberg & Lee, arxiv.org/abs/1705.07874 · attention-weight attribution
- More datasets — UNSW-NB15 · CTU-13 · CICIoT2023 · LANL auth (loaders in [ml/world_model/dataset_loader.py](#))
- Reproducible baseline harness — World Model vs Logistic Regression: [ml/world_model/benchmark.py](#)